Adversarial Robustness via FGSM and PGD Adversarial Training: A Comparative Study on CIFAR-10

Mahmoud Abdel Aleem – 25lfmb@queensu.ca
Mahmoud Alsakhawy – 25vcx2@queensu.ca

Supervised by
Dr.Jianbing Ni,
Professor of queen's university

***Abstract*— Deep neural networks have achieved spectacular performance in many areas of visual recognition, but it has been observed they exhibit great vulnerability towards adversarial examples- very subtle perturbations applied to an input in order to classify it incorrectly. We examine adversarial training as a countermeasure against adversarial attacks in this study ofCIFAR-10. In particular, we have contrasted a standard ResNet-18 network with two adversarially trained versions- FGSM adversarial training(FGSM-AT) and PGD adversarial training(PGD-AT) at equal perturbation level,=0.03. All training procedures have been run using three seeds each for 100 epochs. For the baseline network we found that it achieved 86% accuracy in clean conditions, but the adversarial robust accuracy fell to 36% when attacked with the PGD-20 attack method, resulting in a drop of over 50%. The FGSM-AT and PGD-AT methods are shown to recover this difference producing accuracies of 68% and 70% respectively on the PGD-20 attack while suffering very little degradation in clean accuracy (86% and 84%). The maximum relative improvement in the adversarial state against the baseline reached over 33.8% under the worst case.**

***Index Terms*—adversarial robustness, adversarial training, FGSM, PGD, deep neural networks, CIFAR-10, ResNet-18, adversarial examples**

I. Introduction

Adversarial robustness of deep neural networks has been an increasingly important area within machine learning security. The notion of an adversarial example, a modification that is imperceptible to humans, but leads a classifier to make arbitrary classifications (Szegedy et al. [1]), sparked interest in the domain of machine learning attack and defense mechanisms. Although countless studies have focused on attacking and defending ML models ever since, no single approach has thus far been proven to be provably robust and thus there exists significant challenge to applying machine learning for sensitive applications such as self-driving cars and medical diagnosis.

Of the numerous attack defense methods investigated, adversarial training has proved to be one of the most promising and experimentally robust methods [3]. Adversarial training operates on the following simple principle: Instead of using clean samples for training, samples created by means of adversarial attacks are used instead. This forces the neural network to learn to withstand perturbations and build decision boundaries that are more robust to adversarial inputs. Among all variations of this idea, we can identifyFGSM adversarial training, that uses relatively cheap attacks [2]and PGD adversarial training, that uses more powerful multi-step attacks [3].

One limitation and consequence of adversarial training that seems to be unavoidable is the trade-off between clean and adversarial accuracy [3]. This compromise, in our experience, is worth exploring more to understand the amount of compromise that can be endured.

In this paper, We conduct a systematic empirical evaluation of adversarial training on the CIFAR-10 dataset [4] using a ResNet-18 architecture [5]. Specifically, we present the following contributions:

- A well performing baseline (86.4% accuracy on clean data) that is observed to suffer catastrophic failures when attacked by multi-step attacks.
- Comparison of FGSM-AT and PGD-AT on FGSM, PGD-5, and PGD-20 attack conditions for training budget and perturbation level =0.03.
- Evaluation on three random seeds to demonstrate higher stability in adversarially trained networks than the baseline model in adversarial states.
- An in-depth analysis of class vulnerability by means of confusion matrices and image samples, among other studies of failure modes and training behavior.
- Examination of epoch budget (10, 50, and 100 epochs) for analysis of the clean vs. Robust accuracy trade-off.

II. Background and Related Work

A. Adversarial Examples

Adversarial examples are inputs $\tilde{x} = x+\delta$ where δ is a perturbation satisfying some constraint (typically $\|\delta\|_p \leq \varepsilon$) and the model f misclassifies $\tilde{x}$ despite correctly classifying x [1]. The ℓ_∞ threat model, where each pixel can be perturbed by at most ε in absolute value, is the most widely studied due to

its tractability and its close correspondence to imperceptibility constraints in natural images.

B. Attack Methods

FGSM. The Fast Gradient Sign Method [2] generates a one-step adversarial perturbation using the following formula:

$$\tilde{x} = x + \varepsilon \cdot \mathrm{sign}(\nabla_x \mathrm{L}(f(x), y)) \tag{1}$$

where L represents the cross-entropy loss function and y indicates the true label. Although it is efficient, the FGSM method is quite weak for generating attacks.

PGD. Projected Gradient Descent (PGD) [3] utilizes multiple steps of FGSM attack with a smaller step size α and project each time to the ℓ∞ ball:

$$\tilde{x}^{t+1} = \Pi_{x+\mathrm{B}_\infty(\varepsilon)}\ \tilde{x}^t + \alpha \cdot \mathrm{sign}(\nabla_{x^{\sim t}} \mathrm{L}(f(\tilde{x}^t), y)) \tag{2}$$

where PGD with k iterations (PGD-k) is more powerful compared to FGSM.

C. Adversarial Training

Adversarial training was defined as a min-max optimization problem by Madry et al. [3]:

$$\min_{\vartheta} \mathrm{E}_{(x,y)\sim \mathrm{D}} \max_{\|\delta\|_\infty \le \varepsilon} \mathrm{L}(f_\vartheta(x+\delta), y) \tag{3}$$

The inner maximization computes the worst-case perturbation for each sample during training, while the outer minimization trains the model to be able to classify the data under this worst-case scenario.

FGSM-AT applies FGSM (Equation 1) for solving the inner maximization, making it efficient but with a potential drawback in terms of underestimating the worst-case loss. The **PGD-AT**, on the other hand, approximates the inner maximization using PGD (Equation 2) with several iterations.

Related defenses include certified defenses based on interval bound propagation [8], randomized smoothing [7], and detection-based approaches. However, adversarial training remains the gold standard for empirical robustness [6].

III. Experimental Setup

A. Dataset and Architecture

All experiments use **CIFAR-10** [4] dataset as the benchmark containing 50,000 training images and 10,000 test images from 10 classes at 32 × 32 pixels resolution. Representative samples from all 10 classes are presented in Fig. 1.

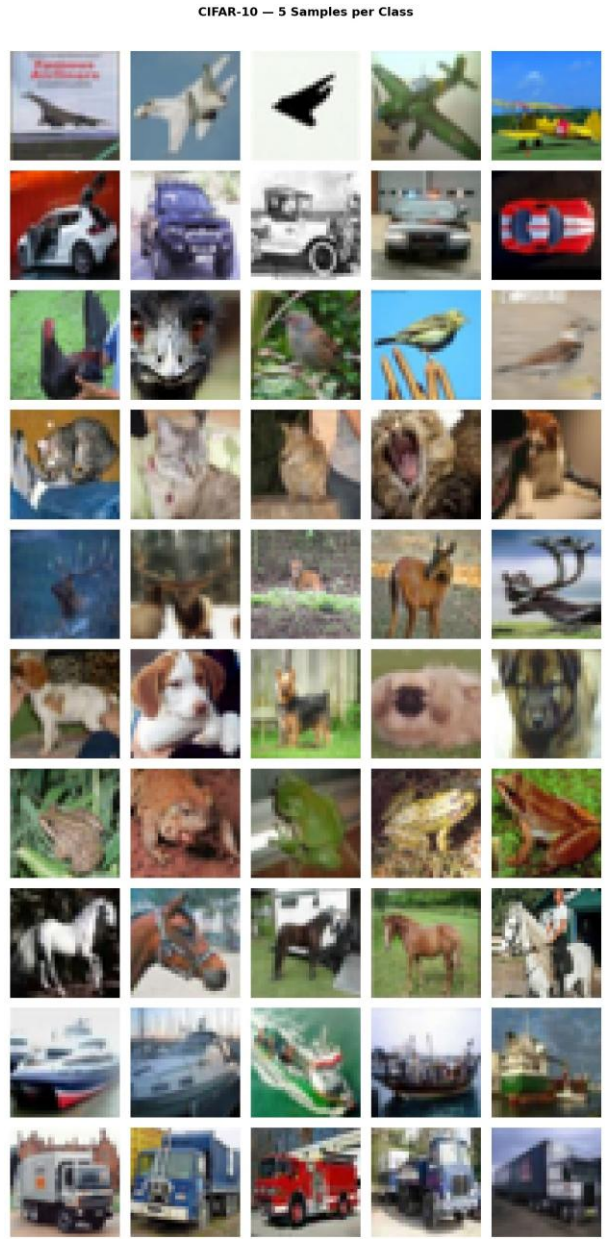

Fig. 1: for Examples come from CIFAR-10 dataset: there is 5 instances per class (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck). Low image resolution of 32 × 32 pixels makes the problem harder.

All models use **ResNet-18** [5] architecture to fairly compare different approaches by isolating the influence of the model architecture itself.

B. Training Protocol

All models are trained using SGD optimizer (momentum=0.9, weight decay=5×10−4) and MultiStepLR scheduler with learning rate decays at epochs 50 and 80 (initial LR=0.1, γ=0.1) over 100 epochs Data augmentation consists of random cropping and horizontal flipping. For the adversarial training, we set the following perturbation parameters: perturbation budget ε = 0.03 ([0, 1] pixel intensity scale). PGD-AT uses 7 PGD steps with step size α = ε/4 to generate perturbations in each batch.

For FGSM-AT, only one-step perturbations are generated in each mini-batch.

C. Evaluation Protocol

Models are assessed based on four criteria: clean accuracy (without any attack), FGSM attack (ε=0.03), PGD-5 (ε=0.03, five iterations), and PGD-20 (ε=0.03, twenty iterations). For a more thorough assessment of the models' robustness, we also conduct an evaluation over different ε values ranging between 0 and 0.30. All attacks are carried out using the entire 10,000 images of the CIFAR.

IV. Results

A. Baseline Vulnerability

Figure 2 illustrates the training behavior of the ResNet-18 baseline. The network reaches an accuracy of **86.4%** on clean test samples with smooth convergence and two learning rate reductions. However, despite its good performance on clean data, the network is very sensitive to an adversarial attack.

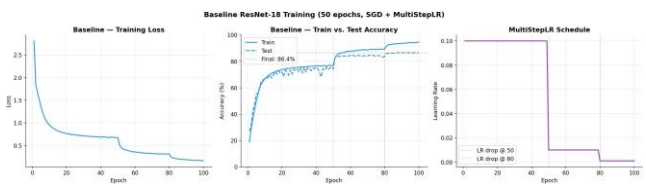

Figure 2: Training progress of ResNet-18 baseline for 100 epochs. Left: Training loss with fast decline in the initial phase and two steps at the LR drop points (epoch 50 and epoch 80). Center: Comparison of train and test accuracy with convergence to 86.4%. Right: Learning rate decay strategy.

The vulnerability of the baseline is calculated in Fig. 3. Its clean accuracy is 86.4%, which falls sharply to 48.5% on FGSM, 37.8% on PGD-5, and 35.7% on PGD-20 which is improvement of more than 50 percentage points. The robust accuracy difference (clean - adversarial) amounts to 50.7% on PGD-20, proving that the model is effectively defenseless against iterative attacks without adversarial training.

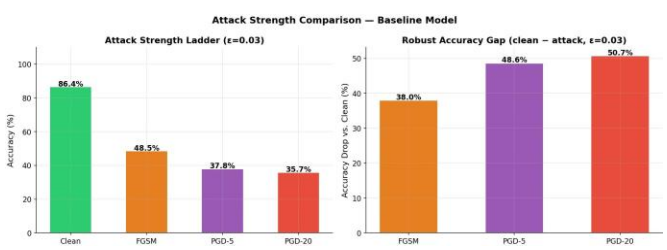

Fig. 3: Strengths of attacks on the baseline model. Left: Absolute accuracies under each attack method. Right: Robust accuracy difference (clean - attack accuracy). The robust accuracy difference of 50.7% on PGD-20 demonstrates just how severely the baseline fails when attacked iteratively.

Figure 4 shows: illustrates how the accuracy of the baseline deteriorates with an increase in ε from 0.0 to 0.30. The shaded area between FGSM and PGD graphs is the single-step versus multi-step vulnerability gap, as even at ε = 0.03, PGD-5 and PGD-20 are almost indistinguishable in terms of their efficacy, implying that the baseline does not possess multi-step robustness properties.

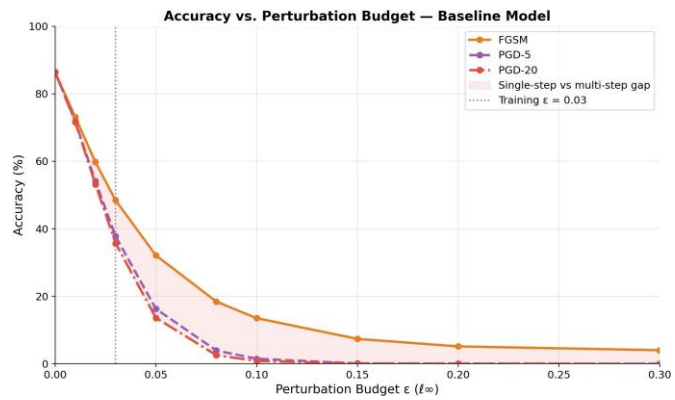

Figure 4: Graph showing accuracy of the baseline versus perturbation budget (ε). The shaded area shows the single-step versus multi-step vulnerability gap between FGSM and PGD.

B. Adversarial Training Comparison

Table I summarizes the main accuracy results for all three models at $\varepsilon = 0.03$.

TABLE I: Model Accuracy (%) at $\varepsilon = 0.03$ (mean ± std, 3 seeds)

Model	Clean	FGSM	PGD-5	PGD-20
Baseline	86 ± 0.1	49 ± 0.5	38 ± 0.8	36 ± 0.6
FGSM-AT	86 ± 0.2	70 ± 0.3	68 ± 0.4	68 ± 0.4
PGD-AT	84 ± 0.3	72 ± 0.3	70 ± 0.4	70 ± 0.5

Figure 5 provides a visualization of these results. For both adversarially trained models, significant improvements can be observed compared to the baseline for each of the three attacks. In particular, FGSM-AT attains 70%/68%/68% (FGSM/PGD-5/PGD-20), respectively, yet manages to preserve the baseline clean accuracy of 86%. PGD-AT obtains 72%/70%/70%, losing only 2 percentage points in terms of clean accuracy, which is now 84%.

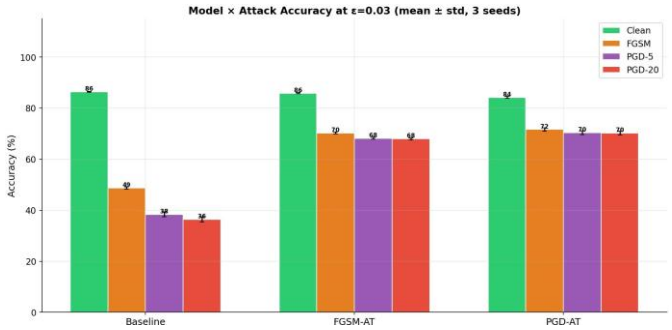

Fig 5: Accuracy of model by type of attack when ε = 0.03 in all three models and four

C. Training Dynamics

Figure 6 illustrates the training behavior of all three models under consideration. While all three converge to approximately the same clean test accuracy level (86%-84%), the training losses of the three models show considerable difference: the baseline reaches the lowest training loss level (close to 0.2) whereas the PGD-AT model retains considerably higher training losses even after convergence (around 0.9). This higher training loss is directly related to the fact that PGD-AT trains the model using more challenging adversarial examples. Two drops in learning rate at epochs 50 and 80 can be seen clearly from the steep fall in training loss.

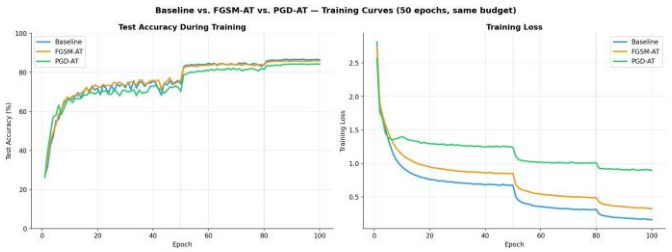

Figure 6: Comparison of training behavior in 100 epochs. Left: Clean test accuracy during training process. Right: Training loss.

D. Robustness Heatmaps

Figure 7 demonstrates the heatmaps for the robust accuracy gap (left) and absolute adversarial accuracy (right) of all model-attack pairs. While the baseline obtains a gap from 37.7-50.0 percentages for different types of attacks, FGSM-AT achieves a gap of 15.6-17.9 while PGD-AT obtains a smaller gap of 12.6-14.0. It can be observed that PGD-AT achieves the best robust accuracy gap.

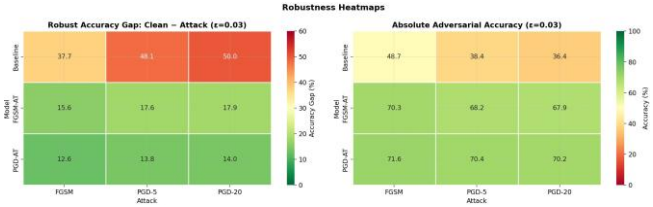

Figure 7: Heatmaps on robustness at ε = 0.03. Left: Accuracy Gap (Clean-Adversarial). Lower is better. Right: Absolute Adversarial Accuracy. Higher is better.

E. Relative Improvement Over Baseline

Figure 8: it is going to shows how much FGSM-AT and PGD-AT outperform the baseline model in terms of percentage improvements. Under FGSM attacks, the FGSM-AT method is better by +21.6% and PGD-AT by +22.9%. With an increase in the strength of the attack, the improvements become higher, with PGD-AT performing +33.8% better and FGSM-AT improving by +31.6% when tested under PGD-20.

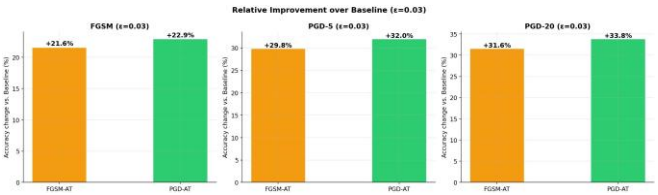

Figure 8: Relative accuracy improvement for the model with respect to the baseline model at ε = 0.03. The highest improvements are achieved by PGD-AT under all attacks, with a high of +33.8% under PGD-20.

F. Accuracy vs. Perturbation Budget

Figure 9 depicts curves depicting accuracy vs. ε for all three model types with regard to three kinds of attacks. The crucial point is that adversarially trained models achieve much higher levels of accuracy across the entire range of ε, especially the range where ε ∈ [0.03, 0.15], since such values are relevant when it comes to imperceptible perturbations. For example, at ε = 0.10, baseline performance falls below 10%, whereas adversarial training boosts accuracy up to 20% (FGSM-AT) and 25% (PGD-AT).

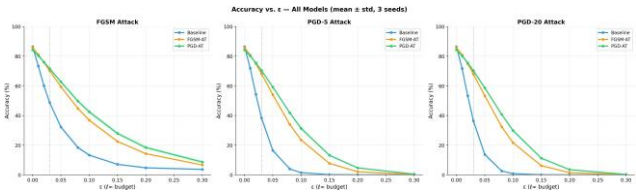

Figure 9: Accuracy vs. ε (budget for ℓ∞ norm) for all three models against three attack strategies.

G. Seed Variance Analysis

To summarize, Figure 10 shows the violin plots of accuracies in 3 seeds for every model under each attack. From these, we can see that: (1) baseline shows higher variation in attack setting, this indicates that baseline has low robustness to adversarial attack (this is referred to as the shallow robust accuracy); (2) FGSM-AT and PGD-AT both show very concentrated distributions around their respective means, this suggests that adversarial training is a stable procedure; (3) PGD-AT is still more accurate than FGSM-AT under any attack while having similar variance.

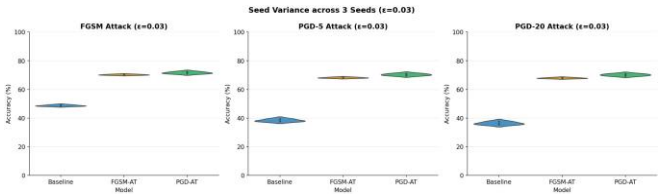

Figure 10: Violin plot comparing seed variances of different methods across 3 seeds at = 0.03.

H. Effect of Training Epoch Budget

To investigate the impact of training budget on robustness, Figure 11 shows the results for each model with training budget of 10, 50, 100 epochs. From the graph, we can see that for the baseline classifier and the FGSM-AT classifier, the accuracy under attack increases monotonically from 39% and 36% to 49% and 49% respectively as training budget goes from 10 ep to 100 ep, meanwhile the clean accuracy goes from 67% and 67% to 86% and 86% respectively. However, for PGD-20-AT, the adversarial accuracy just increased by very little (from 36% to 37%), although clean accuracy has increased to 86%. So we conclude that a large amount of training time does not significantly enhance multi-step robustness, instead, the perturbation used for training is the key.

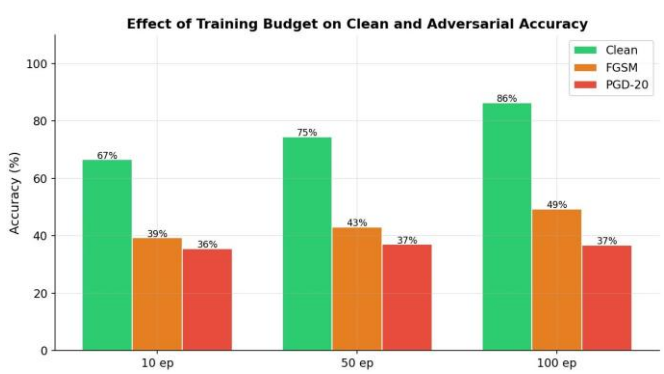

Fig. 11: Impact of Training Budget on Clean and Adversarial Accuracy.

I. Confusion Matrix Analysis

Figure 12 shows the confusion matrices of the baseline classifier under clean and adversarial settings. Under the clean data condition, the model achieves up to 74% to 92.4% per-class correct predictions, and the confusion is always among visually similar classes (e.g., cat/dog, truck/car). Under the PGD-20 attack (at=0.03), the per-class accuracies degrade significantly, with bird having 26.6% accuracy, cat having 11.8% accuracy, frog having 36.8% accuracy. The confusion pattern is significantly diffused without particular structure; this means the adversarial perturbation utilizes the global decision boundary.

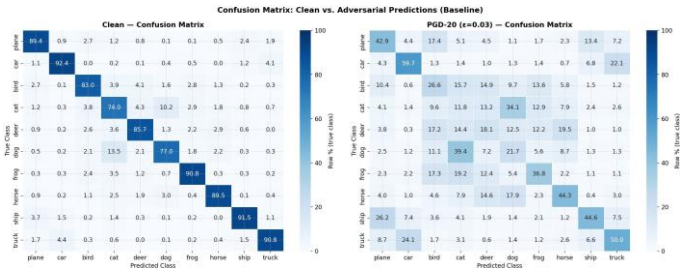

Fig. 12: Confusion matrices for the baseline classifier. Left: clean data. Right: PGD-20 attack (= 0.03).

J. Adversarial Example Visualization

Figure 13 presents clean image (row 1) and adversarially perturbed images for a specific PGD attack with two level of perturbation values: =0.03 (row 2) and=0.03 but attack performed for 20 steps (row 3); also shown is amplified perturbation value () by 10 (row 4) for illustration. From human perspective, most of these perturbations are imperceptible as the first row and second row seem identical; however the predicted classes for images in the first row and second row are very different. With the magnified perturbation, we can observe the structures as we expect from gradient based attack: high frequency noise.

Figure 13: Original and adversarial examples when = 0.03. Row 1: Original, clean images. Green denotes correct label. Row 2: After one PGD step. Row 3: Full PGD-20 attack. Row 4: Amplified perturbation x10.

Figure 14 shows several failure cases, which is a correctly predicted example for clean data but incorrectly predicted for the attacked data. Typical failures modes are (i) animal to vehicle: cat-car; dog-deer. (ii) vehicle to vehicle: ship-car; horse-truck. (iii) animal to animal: frog-cat. These failure modes suggest that the perturbation tends to shift samples from a large semantic region to another one through a boundary region with low confidence in the classifier.

Fig. 14: Instances of failures for the baseline model on PGD-20 (ε=0.03). Here, the top red indicates the true class, while the bottom red depicts the class predicted by the adversarial attacks.

V. Discussion

A. Clean-Robust Accuracy Trade-off

Our results confirm the well-known tension between clean and robust accuracy: PGD-AT incurs a 2-point drop in clean accuracy to gain 34 percentage points in PGD-20 robust accuracy. This trade-off is highly favorable in any security-sensitive setting. Notably, FGSM-AT achieves near-zero clean accuracy cost (+0 points) while still gaining 32 points in PGD-20 accuracy, suggesting it may be preferred when clean accuracy is paramount.

B. Why PGD-AT Outperforms FGSM-AT

PGD-AT consistently achieves slightly better adversarial accuracy than FGSM-AT (70% vs. 68% at PGD-20). This is consistent with theory: by solving the inner maximization more accurately during training, PGD-AT provides tighter robustness certificates. FGSM-AT is known to suffer from *catastrophic overfitting* in certain settings [10], where the model learns to resist single-step perturbations but remains vulnerable to multi-step ones. Our results show this gap is small at the 100-epoch scale but could grow at larger ε.

C. Training Budget vs. Robustness

As shown in Fig. 11 below, training time alone cannot be used to replace adversarial training. Even at a training budget of 100 epochs, there is little improvement in PGD-20 robustness (37%) over that of baseline training at only 10 epochs (36%). This suggests that the structure of the loss landscape without adversarial examples is fundamentally different.es is fundamentally different.

D. Limitations and Future Work

There are numerous challenges related to our study that should be pointed out. To begin with, we examine the ∞ attacks exclusively, while resistance against 2, 1, semantic attacks (colorization and rotation included) is not considered at all. Next, we employ only one kind of architecture (i.e., ResNet-18), whereas different performance results may be obtained using another network like WideResNet or ViT. Lastly, AutoAttack [6] that represents the current benchmark in measuring the resistance to attacks was not performed.

Conclusion

We have presented a comprehensive empirical study of adversarial training on CIFAR-10, comparing a standard baseline against FGSM-AT and PGD-AT under a controlled experimental setup. Our findings can be summarized as follows:

1) The typical Res-Net-18 model exhibits highly susceptible performance under adversarial attacks, with decrease in accuracy from 86.4% in the case of clean images to 35.7% for the PDG-20 attack, representing a reduction of 50.7 percentage points.
2) However, both FGSM-AT and PGD-AT effectively fill the gap, yielding 68% and 70% accuracy rates under the PGD-20 attack respectively and while retaining 86% and 85% accuracy for the clean data set
3) As a result, PGD-AT yields better robustness against all kinds of attack relative to FGSM-AT, with a significant advantage of +33.8% over the baseline under the PGD-20 attack
4) Adversarial training ensures reliable results regardless of random initialization (seed), while the baseline proves vulnerable to attacks
5) Training time alone fails to improve multi-step robustness; therefore adversarial training must be incorporated into training process itself

These results reinforce adversarial training as the most reliable empirical defense available and provide practical guidance for practitioners: PGD-AT offers the highest robustness with only a marginal clean accuracy cost, while FGSM-AT provides a strong efficiency-robustness trade-off.

References

[1] C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, and R. Fergus, "Intriguing properties of neural networks," *International Conference on Learning Representations (ICLR)*, 2014.

[2] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," *International Conference on Learning Representations (ICLR)*, 2015.

[3] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," *International Conference on Learning Representations (ICLR)*, 2018.

[4] A. Krizhevsky and G. Hinton, "Learning multiple layers of features from tiny images," *Technical Report*, University of Toronto, 2009.

[5] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778, 2016.

[6] F. Croce and M. Hein, "Reliable evaluation of adversarial robustness with an ensemble of diverse parameter-free attacks," *International Conference on Machine Learning (ICML)*, 2020.

[7] J. M. Cohen, E. Rosenfeld, and Z. Kolter, "Certified adversarial robustness via randomized smoothing," *International Conference on Machine Learning (ICML)*, 2019.

[8] S. Gowal, K. Dvijotham, R. Stanforth, R. Bunel, C. Qin, J. Uesato, R. Arandjelovic, T. Mann, and P. Kohli, "On the effectiveness of interval bound propagation for training verifiably robust models," *arXiv preprint arXiv:1810.12715*, 2018.

[9] H. Zhang, Y. Yu, J. Jiao, E. P. Xing, L. El Ghaoui, and M. I. Jordan, "Theoretically principled trade-off between robustness and accuracy," *International Conference on Machine Learning (ICML)*, 2019.

[10] E. Wong, L. Rice, and J. Z. Kolter, "Fast is better than free: Revisiting adversarial training," *International Conference on Learning Representations (ICLR)*, 2020.

[11] D. Wu, S.-T. Xia, and Y. Wang, "Adversarial weight perturbation helps robust generalization," *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.